

JEE MAIN REVISION MASTER: ALTERNATING CURRENT

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 191 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Series LCR Circuit & Impedance	85 PYQs	Circuits
Very High	AC Fundamentals: Peak, RMS & Phase	74 PYQs	Foundations
Very High	Reactance: Inductive & Capacitive	68 PYQs	Impedance Elements
Very High	Resonance, Q-Factor & Bandwidth	63 PYQs	Resonance
Very High	Power in AC Circuits & Power Factor	40 PYQs	Power & Efficiency
Very High	LC Oscillations & Transformers	24 PYQs	Devices & Oscillations

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Series LCR Circuit & Impedance	$Z = \sqrt{R^2 + (X_L - X_C)^2}$, $\tan\phi = \frac{X_L - X_C}{R}$, $V_{\text{total}} = \sqrt{V_R^2 + (V_L - V_C)^2}$
AC Fundamentals: Peak, RMS & Phase	$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$, $V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$, $I_{\text{avg, half}} = \frac{2 I_0}{\pi}$
Reactance: Inductive & Capacitive	$X_L = \omega L = 2\pi f L$, $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$
Resonance, Q-Factor & Bandwidth	$\omega_0 = \frac{1}{\sqrt{LC}}$, $Z_{\min} = R$, $I_{\max} = \frac{V}{R}$, $Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{\omega_0}{\Delta\omega}$
Power in AC Circuits & Power Factor	$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\phi$, $\cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$, $I_{\text{wattless}} = I_{\text{rms}} \sin\phi$
LC Oscillations & Transformers	$V_s V_p = V_{N_s} V_{N_p} = I_p I_s$, $\eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{V_s I_s}{V_p I_p}$, $q(t) = q_0 \cos(\omega_0 t)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Series LCR Circuit & Impedance (Very High Yield — 85 PYQs)

Core Principle: Series LCR impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$. Circuit current $I = (V / Z)$. Phase angle $\tan\phi = (X_L - X_C / R)$. Net voltage phasor $V_{\text{(total)}} = \sqrt{V_R^2 + (V_L - V_C)^2}$.

■ **Caution / JEE Pitfall:** Voltages across series AC elements do NOT add algebraically: $V_{\text{(total)}} \neq V_R + V_L + V_C$! Always use phasor vector addition: $V_{\text{(total)}} = \sqrt{V_R^2 + (V_L - V_C)^2}$.

■ **Speed Trick:** If $X_L > X_C$, circuit is inductive (voltage leads current by ϕ). If $X_C > X_L$, circuit is capacitive (current leads voltage by ϕ).

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A LCR series circuit driven with $E_{\text{(rms)}}=90\text{V}$ at frequency $f_{\text{(d)}}=30\text{Hz}$ has resistance $R=80\Omega$, an inductance with inductive reactanc..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "For the series LCR circuit connected with $220\text{V}, 50\text{Hz}$ a.c source as shown in the figure, the power factor is $(\alpha / 10)$. The value..." [Ans: C]
- [2026 Evening Shift] Q#1 (Numerical): "A series LCR circuit with $R=20\Omega, L=1.6\text{H}$ and $C=40\mu\text{F}$ is connected to a variable frequency a.c. source. The inductive reactance at $r...$ " [Ans: 200]

3.2 AC Fundamentals: Peak, RMS & Phase (Very High Yield — 74 PYQs)

Core Principle: Sinusoidal AC quantities: $I_{\text{(rms)}} = \frac{I_0}{\sqrt{2}} \approx 0.707 I_0$, $V_{\text{(rms)}} = \frac{V_0}{\sqrt{2}}$. Average current over full cycle is 0; average over half cycle is $I_{\text{(avg, half)}} = (2I_0 / \pi) \approx 0.637 I_0$.

■ **Caution / JEE Pitfall:** AC voltmeters and ammeters measure RMS values, not peak values! When JEE specifies '220 V AC', it always denotes $V_{\text{(rms)}}$ (peak voltage is $V_0 = 220\sqrt{2} \approx 311\text{ V}$).

■ **Speed Trick:** Instantaneous power $p(t) = v(t)i(t)$ fluctuates at double frequency (2ω) about average power $P_{\text{(avg)}} = V_{\text{(rms)}}I_{\text{(rms)}}\cos\phi$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A LCR series circuit driven with $E_{\text{(rms)}}=90\text{V}$ at frequency $f_{\text{(d)}}=30\text{Hz}$ has resistance $R=80\Omega$, an inductance with inductive reactanc..." [Ans: A]
- [2026 Evening Shift] Q#3 (MCQ): "The figure given below shows an LCR series circuit with two switches S_1 and S_2 . When switch S_1 is closed keeping S_2 open, the phase dif..." [Ans: B]
- [2026 Evening Shift] Q#3 (Numerical): "An inductor stores 16 J of magnetic field energy and dissipates 32 W of thermal energy due to its resistance when an a.c. current of 2 A (rm..." [Ans: 314]

3.3 Reactance: Inductive & Capacitive (Very High Yield — 68 PYQs)

Core Principle: Pure Inductor: Voltage leads current by 90° ($(\pi/2)$), inductive reactance $X_L = \omega L = 2\pi f L$. Pure Capacitor: Current leads voltage by 90° , capacitive reactance $X_C = (1/\omega C) = (1/2\pi f C)$.

■■ **Caution / JEE Pitfall:** In direct current (DC, $f = 0$), an inductor offers zero reactance ($X_L = 0$, short circuit) while a capacitor offers infinite reactance ($X_C \rightarrow \infty$, blocking DC completely).

■ **Speed Trick:** Mnemonic 'CIVIL': In a Capacitor (C), Current (I) leads Voltage (V); in an Inductor (L), Voltage (V) leads Current (I).

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A LCR series circuit driven with $E_{(rms)}=90V$ at frequency $f_{(d)}=30Hz$ has resistance $R=80\Omega$, an inductance with inductive reactanc..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "For the series LCR circuit connected with 220V,50Hz a.c source as shown in the figure, the power factor is $(\alpha/10)$. The value..." [Ans: C]
- [2026 Evening Shift] Q#1 (Numerical): "A series LCR circuit with $R=20\Omega, L=1.6H$ and $C=40\mu F$ is connected to a variable frequency a.c. source. The inductive reactance at r..." [Ans: 200]

3.4 Resonance, Q-Factor & Bandwidth (Very High Yield — 63 PYQs)

Core Principle: At resonance: $X_L = X_C$, resonant frequency $\omega_0 = \frac{1}{\sqrt{LC}}$ ($f_0 = \frac{1}{2\pi\sqrt{LC}}$). Minimum impedance $Z_{(min)} = R$, maximum current $I_{(max)} = (V/R)$. Quality factor $Q = (\omega_0 L / R) = (1 / \omega_0 C R) = (1/R)\sqrt{(L/C)} = (\omega_0 / \Delta\omega)$.

■■ **Caution / JEE Pitfall:** At resonance, V_L and V_C are NOT zero! In fact, $V_L = V_C = Q \cdot V_{(supply)}$, which can be substantially higher than supply voltage (voltage magnification)!

■ **Speed Trick:** Resonance bandwidth $\Delta\omega = (R/L)$. To sharpen resonance (higher Q), decrease resistance R or increase the (L/C) ratio.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (Numerical): "A series LCR circuit with $R=20\Omega, L=1.6H$ and $C=40\mu F$ is connected to a variable frequency a.c. source. The inductive reactance at r..." [Ans: 200]
- [2026 Evening Shift] Q#2 (MCQ): "A capacitor C is first charged fully with potential difference of $V_{(0)}$ and disconnected from the battery. The charged capacitor is connec..." [Ans: N/A]
- [2026 Morning Shift] Q#4 (Numerical): "Using a variable frequency a.c. voltage source the maximum current measured in the given LCR circuit is 50 mA for $V=5\sin(100t)$ The values ..." [Ans: 50]

3.5 Power in AC Circuits & Power Factor (Very High Yield — 40 PYQs)

Core Principle: Average power dissipated $P_{\text{(avg)}} = V_{\text{(rms)}} I_{\text{(rms)}} \cos\phi$. Power factor $\cos\phi = (R / Z) = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$. Wattless current component $I_{\text{(wattless)}} = I_{\text{(rms)}} \sin\phi$.

■ **Caution / JEE Pitfall:** Power factor is $\cos\phi = (R / Z)$, NOT $\tan\phi$ or (R / X) ! For pure L or pure C, $\phi = 90^\circ$ implies $\cos\phi = 0$, so average power dissipation is strictly zero.

■ **Speed Trick:** Apparent power $S = V_{\text{(rms)}} I_{\text{(rms)}} (in VA)$. True power $P = S \cos\phi (in W)$. Reactive power $Q = S \sin\phi (in VAR)$.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "A LCR series circuit driven with $E_{\text{(rms)}}=90V$ at frequency $f_{\text{(d)}}=30Hz$ has resistance $R=80\Omega$, an inductance with inductive reactanc..." **[Ans: A]**
- **[2026 Morning Shift] Q#1 (MCQ):** "For the series LCR circuit connected with 220V,50Hz a.c source as shown in the figure, the power factor is $(\alpha / 10)$. The value..." **[Ans: C]**
- **[2026 Morning Shift] Q#2 (Numerical):** "An inductor of 10 mH, capacitor of $0.1\mu F$ and a resistor of 100Ω are connected in series across an a.c power supply $220V, 70Hz$..." **[Ans: 100]**

3.6 LC Oscillations & Transformers (Very High Yield — 24 PYQs)

Core Principle: Ideal transformer equations: $(V_s / V_p) = (N_s / N_p) = (I_p / I_s)$. Efficiency $\eta = \frac{P_{\text{(out)}}}{P_{\text{(in)}}} = \frac{V_s I_s}{V_p I_p}$. LC tank oscillations: charge $q(t) = q_0 \cos(\omega_0 t)$ with total energy conserved $U = (q^2 / 2C) + (1 / 2) L I^2$.

■ **Caution / JEE Pitfall:** Transformers operate exclusively on AC; in DC, primary flux is static ($(d\Phi / dt) = 0$), producing zero secondary voltage and potentially burning the primary coil!

■ **Speed Trick:** In step-up transformers ($N_s > N_p$), voltage increases but current decreases, keeping total power $V I$ conserved under ideal conditions.

Representative Source PYQs:

- **[2026 Evening Shift] Q#2 (MCQ):** "A capacitor C is first charged fully with potential difference of $V_{\text{(0)}}$ and disconnected from the battery. The charged capacitor is connec..." **[Ans: N/A]**
- **[2024 Morning Shift] Q#1 (MCQ):** "In an ac circuit, the instantaneous current is zero, when the instantaneous voltage is maximum. In this case, the source may be connected to..." **[Ans: C]**
- **[2024 Morning Shift] Q#2 (MCQ):** "Primary coil of a transformer is connected to 220V ac. Primary and secondary turns of the transforms are 100 and 10 respectively. Secondar..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

A LCR series circuit driven with $E_{\text{rms}}=90\text{V}$ at frequency $f=30\text{Hz}$ has resistance $R=80\Omega$, an inductance with inductive reactance $X_L=20.0\Omega$ and capacitance with capacitive reactance $X_C=80.0\Omega$. The power factor of the circuit is ____.

- (A) 0.8 **[Correct]**
- (B) 0.64
- (C) 0.9
- (D) 0.5

Correct Answer: A

Solution Analysis: In an AC circuit, the power factor ($\cos\phi$) is defined as the ratio of the resistance (R) to the total impedance (Z) of the circuit : $\text{Power Factor}=\cos\phi=(R/Z)$ For a series LCR circuit, the impedance is $Z=\sqrt{R^2+(X_C-X_L)^2}$ The value of resistance is $R=80\Omega$, the inductive reactance is $X_L=20.0\Omega$, and the capacitive reactance is $X_C=80.0\Omega$. $X_{\text{net}}=|X_L-X_C|=|20-80|=60\Omega$ So, the impedance is, $Z=\sqrt{80^2+60^2}=\sqrt{6400+3600}=\sqrt{10000}=100\Omega$ Thus, the power factor is, $\cos\phi=\frac{R}{Z}=\frac{80}{100}$

Benchmark Problem 2: JEE Main 2026 (Online) 24th January Morning Shift (Q#1)

For the series LCR circuit connected with $220\text{V}, 50\text{Hz}$ a.c source as shown in the figure, the power factor is $(\alpha/10)$. The value of α is ____.

- (A) 4
- (B) 8
- (C) 6 **[Correct]**
- (D) 10

Correct Answer: C

Solution Analysis: The power factor ($\cos\phi$) of an AC circuit is defined as the ratio of the true resistance (R) to the total impedance (Z) of the circuit. $\text{Power Factor}=\cos\phi=(R/Z)$ In a series LCR circuit, the inductive reactance (X_L) and capacitive reactance (X_C) oppose each other. The net reactance is the difference between them. $X=|X_L-X_C|$ Substituting the given values : $X=|70-150|\Omega=80\Omega$ The total impedance of the circuit is, $Z=\sqrt{R^2+X^2}$ $Z=\sqrt{(60)^2+(80)^2}=\sqrt{10000}=100\Omega$ So, the power factor is $\cos\phi=\frac{R}{Z}=\frac{60}{100}=\frac{3}{5}$

Benchmark Problem 3: JEE Main 2025 (Online) 22nd January Evening Shift (Q#1)

A series LCR circuit is connected to an alternating source of emf E . The current amplitude at resonant frequency is I_0 . If the value of resistance R becomes twice of its initial value then amplitude of current at resonance will be

- (A) $\frac{I_0}{2}$ **[Correct]**
- (B) $\frac{I_0}{\sqrt{2}}$
- (C) $2I_0$
- (D) I_0

Correct Answer: A

Solution Analysis: At resonance in a series LCR circuit, the reactive components (inductive and capacitive) cancel each other out, leaving only the resistance. Therefore, the amplitude of the current is given by: $I_0=(E/R)$ Now, if the resistance is doubled, the new resistance becomes $2R$. The current amplitude at resonance then becomes: $I=(E/2R)=(1/2)(E/R)=\frac{I_0}{2}$ Thus, the amplitude of current at resonance when the resistance is doubled is $\frac{I_0}{2}$. The correct option is A.